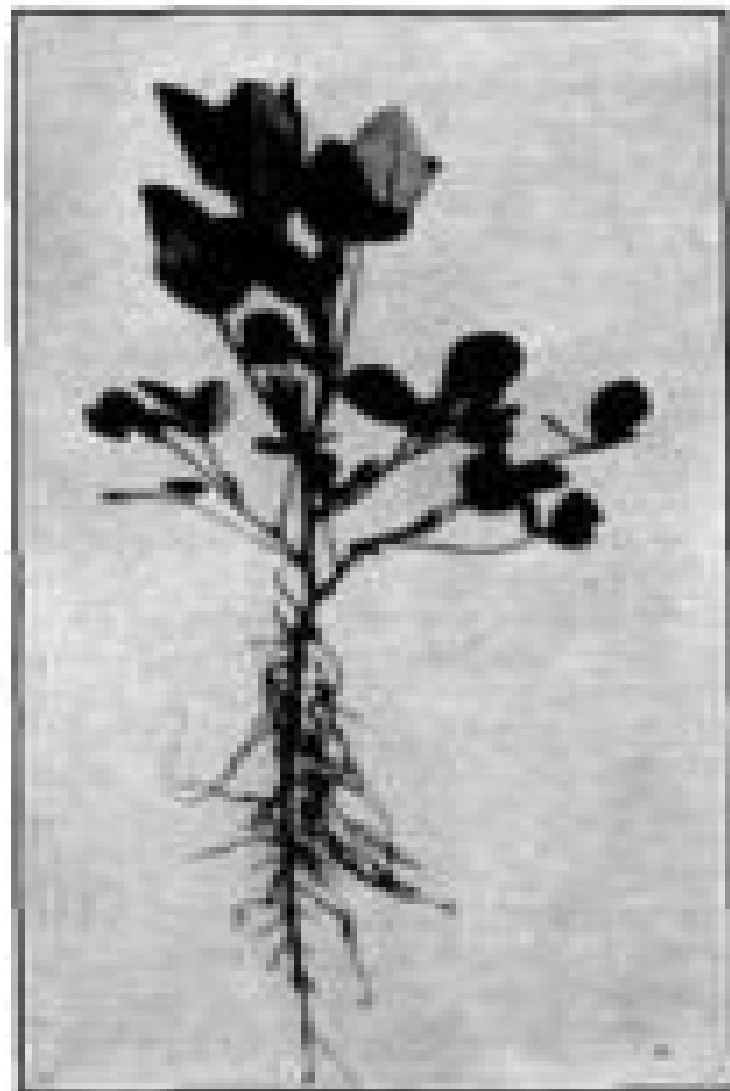


SWEET CLOVER: GROWING THE CROP

H. S. COE

Assistant Agronomist, Office of Foreign-Crop Research



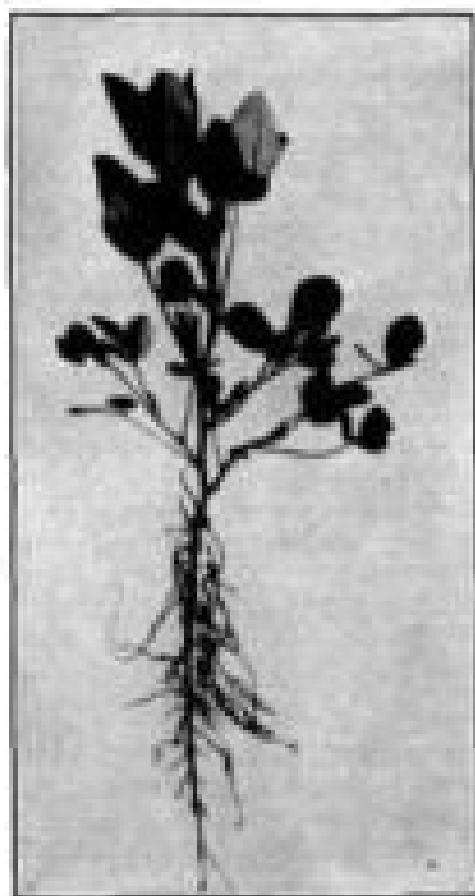
FARMERS' BULLETIN 797
UNITED STATES DEPARTMENT OF AGRICULTURE

Contribution from the Bureau of Plant Industry
WM. A. TAYLOR, Chief

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Washington, D. C.

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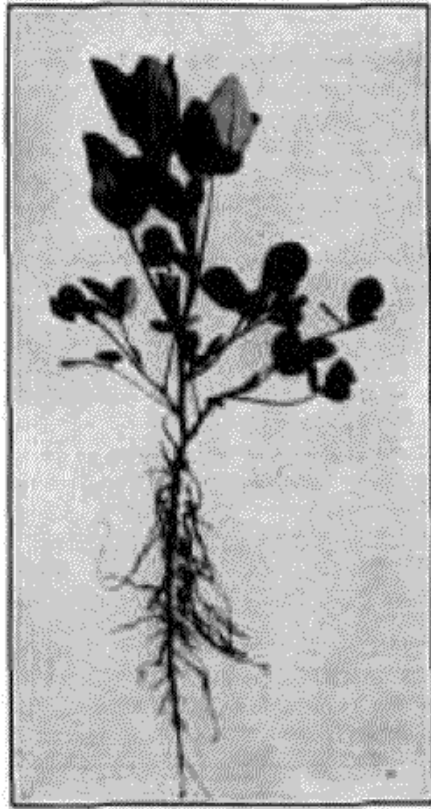
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GROWING THE CROP ***

SWEET CLOVER: GROWING THE CROP

H. S. COE

Assistant, Office of Forage Crop Investigations



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UNITED STATES DEPARTMENT OF AGRICULTURE

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WM. A. TAYLOR, chief

Washington, D. C.

April, 1917

T HE cultivation of sweet clover should be preceded by a through knowledge of the requirements for obtaining a stand.

The white species comprises a very large percentage of the present acreage of sweet clover.

Annual yellow sweet clover should be sown in no portion of the United States except the South and Southwest, and then only as a cover of green-manure crop.

Sweet clover is being cultivated in practically every State in the Union. At the present time the largest acreage is found in The western North-Central States and in the Mountain States.

Sweet clover is adapted to a wider range of climatic conditions than any of the true clovers, and possibly alfalfa.

Sweet clover will grow on practically all soil types to be found in this country, provided the soil is not acid and is well inoculated.

Sweet clover is more drought resistant than alfalfa or red clover. It is quite resistant to alkali.

The lime requirement of sweet clover is as high as that of red clover or alfalfa. Maximum growth is obtained only on soils that are not acid.

Sweet clover usually will respond to applications of fertilizers and manure.

In the more humid sections of the country good stands usually are obtained by seeding with a nurse crop.

Only seed which germinates 75 per cent or more should be sown in the spring of the year unless the rate of seeding is increased to make up for poor germination.

Sweet clover does best when seeded on a well-firmed seed bed which has only sufficient loose soil on the surface to cover the seed.

It is very essential that inoculation be provided in some form if success is to be expected.

The large number of failures in obtaining a stand of sweet clover are due primarily to acid soils, lack of inoculation, and seed which germinates poorly.

Spring seedings in general are satisfactory, but in the South excellent stands are obtained from midwinter seedings also Fall seedings are usually successful south of the latitude of southern Ohio.

A Farmers' Bulletin (No. 820) on the utilization of sweet clover for pasture, hay, and as a green-manure is about to be issued.

SWEET CLOVER: GROWING THE CROP.

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INTRODUCTION.

Sweet clover is an important forage crop in many regions. Although one of the oldest of known plants, not until very recently has it been considered seriously as a forage plant in this country. The principal causes for not utilizing this crop were its aggressiveness on uncultivated land in many localities, the tendency of the stems to become woody as they mature, and the refusal of stock to eat sweet clover before they had become accustomed to the bitter taste. Another reason was the fact that until recently red clover could be grown in the eastern half of the United States without difficulty. In northern Kentucky the continuous growing of tobacco or of tobacco and wheat impoverished the soil to such an extent that crops no longer could be grown successfully. Upon the abandoned farms in this section sweet clover was introduced as a honey plant. Owing to the remarkable yields of tobacco that were obtained on such farms after sweet clover had been grown for a few years the acreage of this plant increased very rapidly. For a number of years sweet clover has been grown on the Selma chalk (rotten-limestone) soils of Alabama and Mississippi as a soil-improving crop. At the present time it is being cultivated in practically every State, and the acreage is increasing very rapidly.

After it had been demonstrated that sweet clover would grow successfully on soils too depleted for other crops, many experiments were conducted to determine its value as forage. It was found that it was not only a valuable soil-improving crop, but that it made an excellent pasture and hay plant, quite palatable and rich in protein.

White sweet clover comprises a very large percentage of the acreage seeded to sweet clover at the present time. On this account this species ordinarily is referred to simply as "sweet clover." The yellow biennial species is designated as yellow sweet clover, and the annual yellow species as bitter clover, sour clover, or annual yellow sweet clover. This usage has been adopted in this bulletin.



FIG. 1.—Seeds and seed pods of three species of *Melilotus* and seeds of alfalfa: 1. White sweet clover; 2, biennial yellow sweet clover; 3, annual yellow sweet clover, or sour clover; 4, alfalfa. The small figures in each drawing show the natural size of the seed. The venation and shape of the seed pods are important characters in distinguishing the different species of sweet clover.

The cultivation of sweet clover should be preceded by a thorough understanding of the requirements for obtaining a stand. It can not be grown successfully on all soils, as many assume from seeing it growing in uncultivated places. Neither will it thrive in many sections of the country without careful preparation of the seed bed. Sweet clover will not grow successfully in acid soils unless lime is applied, but it will make a good growth in soils too low in humus to grow red clover, provided the soil is neutral or alkaline.

Sweet clover is an excellent plant to precede alfalfa, as the large roots do much toward breaking up and aerating the subsoil. Contrary to the belief of many, it will not inoculate the soil for alfalfa unless inoculation is applied to the sweet clover. If, however, the soil contains but few inoculating germs, the sweet clover will serve as a medium to inoculate it thoroughly.

SPECIES OF SWEET CLOVER.

A number of species of sweet clover are found throughout the world, and most of them are native to temperate Europe and Asia as far east as Tibet,

White sweet clover,^[1] yellow biennial sweet clover,^[2] and yellow annual sweet clover^[3] are the only species which have given sufficient promise as forage and green-manure crops in this country to warrant growing them under cultivation.

^[1] *Melilotus alba* Dear.

^[2] *Melilotus officinalis* (L.) Lam.

^[3] *Melilotus indica* (L.) All.

It is difficult for the average person to distinguish between the different species of sweet clover from an examination of the seeds or seed pods only. The differences are indicated in [figure 1](#). Where there is a question as to the identity of a sample of seed it should be sent to a State agricultural experiment station or to one of the seed laboratories of the United States Department of Agriculture for identification.



FIG. 2.—A branch of white sweet clover, showing the long, loose racemes which bear white flowers.

WHITE SWEET CLOVER.

White sweet clover ([fig. 2](#)) is ordinarily referred to as melilotus or meliot in the South and merely as sweet clover in other portions of the country. When soil conditions are favorable for germination, sweet-clover seedlings will appear from one to two weeks after seeding. On account of the biennial nature of the plants, they do not seem to make much growth above ground the first month or six weeks after germination, but during this time they are developing root systems rapidly and thus becoming established, so to be able to withstand adverse conditions. Plants which have made no more than 2 inches of top growth very often have produced roots 6 inches or more in length ([fig. 3](#)). The tap-root continues to develop rapidly throughout the growing

season the first year, and by autumn often reaches a length of 24 to 36 inches and a diameter of three-fourths to 1 inch at the crown.

After the root system becomes established the plants produce an upright, branching, leafy growth, which under ideal growing conditions may reach a height of 48 inches the first season, but more often 18 to 30 inches. A large quantity of reserve food is stored in the tap-root the first season; this reserve food enables the plants to make a rapid and vigorous growth early the following spring.

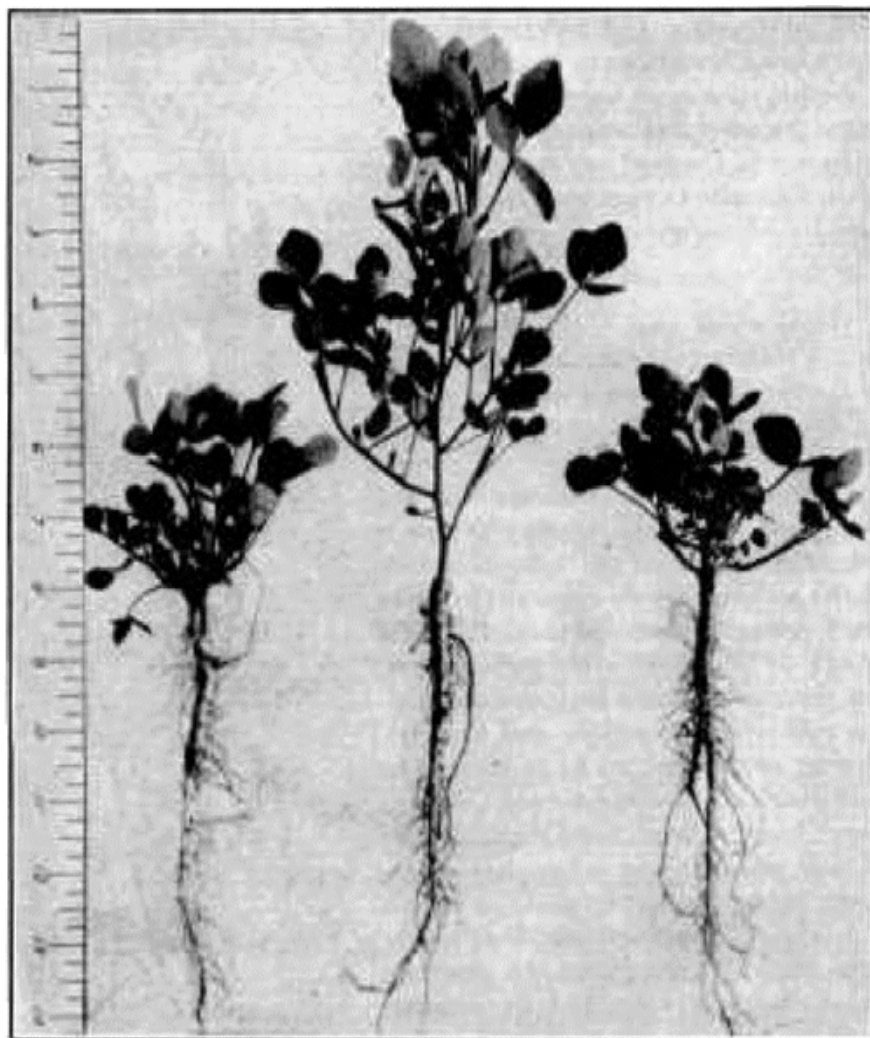


FIG. 3.—White sweet-clover plants collected from a plat six weeks from the date of seeding. An extensive root system such as is shown here is often developed before much growth is made above ground.

Toward the end of the growing season of the first year a number of buds, which serve to produce the second year's growth ([fig. 4](#)), are formed on the crowns of the plants. After these buds are formed the plants may be clipped quite close to the ground, as the buds are not developed until the plants have made sufficient growth to live through the winter.

During the second season sweet clover makes a rapid, erect, stemmy, branching growth from 5 to 10 feet in height, the plants producing only a moderate number of leaves, which drop as the seed matures. A large number of loose racemes bearing white flowers (see [fig. 2](#)) are produced during the flowering period, which usually lasts from three to five weeks.

Before sweet clover has made a growth of 12 to 18 inches it closely resembles alfalfa. The plants may be distinguished from alfalfa by the absence of pubescence on the under side of the leaves and by their bitter taste. When they are in bloom they may be identified easily by their long, loose racemes of white flowers and their open, coarse growth. Unlike alfalfa, the seeds are ordinarily found singly in the pods. Two seeds may occasionally be found, and very rarely three, in a single pod.

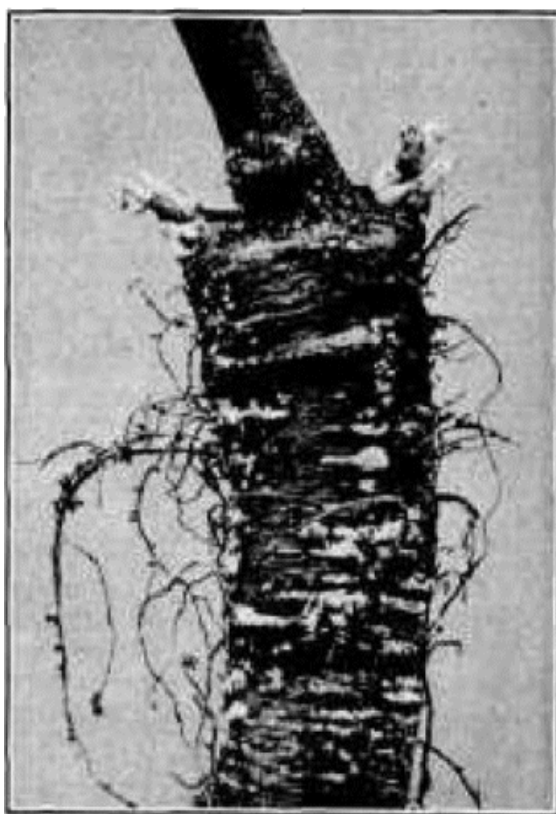


FIG. 4.—Buds produced on the crown of a sweet-clover plant at the end of the first season's growth. These buds will produce the first crop the second season.

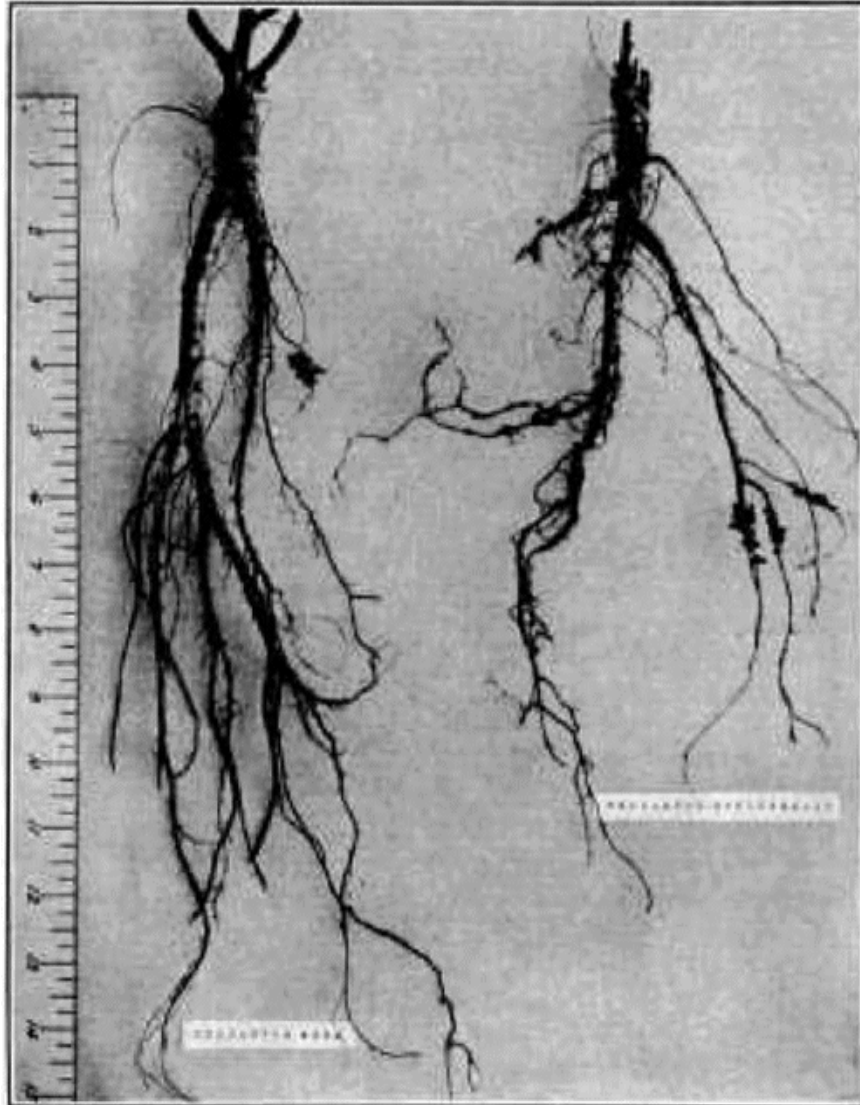


FIG. 5.—Root of white sweet clover (on the left) and of biennial yellow sweet clover (on the right). These roots were collected on October 28, 1915, at Arlington, Va., from adjacent plats seeded to cuts and sweet clover on April 10, 1915. Note the difference in the size of the roots. Tubercles are present on the right-hand side of each root.

STRAINS OF WHITE SWEET CLOVER.

A number of different strains of white sweet clover are to be found in the average field, but most of them are not as marked or as conspicuous as the different strains of red clover. The principal differences between strains of sweet clover are in leafiness, habit of growth, and date of blooming.

Occasional plants are especially heavy seed producers and bear many pods containing more than one seed. Other plants bloom earlier than the average date for

white sweet clover, and it may be possible by selecting such strains to find one which matures early enough to produce two crops a season at high altitudes in the northern sections of the United States.

Fields of an exceptionally early blooming strain were found in Illinois, Iowa, and North Dakota in the summer of 1916. The plants were different in type of growth from the ordinary white sweet clover, being most conspicuous from the fact that they were in bloom during the first week of June, which is at least three weeks earlier than the ordinary species should bloom in these localities.

An annual white-flowered sweet clover was found in several localities in the fall of 1916. The seed which produced these plants was grown in Alabama. These plants resembled *Melilotus alba* in most respects except that they were strictly annual. They flowered and matured seed abundantly in South Dakota and North Dakota. It has not been determined whether this is a distinct species or merely an annual strain of the species mentioned.

BIENNIAL YELLOW SWEET CLOVER.

Biennial yellow sweet clover ordinarily is referred to in the seed trade and among farmers in regions where it is grown simply as yellow sweet clover. The plants of this species are somewhat more decumbent the first year, and ordinarily with more deeply notched leaves than the white-flowering species. Yellow sweet clover usually grows from 3 to 5 feet in height. This plant blooms from 10 to 14 days earlier than the white species, and for this reason it is advisable to sow seed of both plants when they are to be used for bee pasturage. On account of the finer stems of yellow sweet clover it is preferred in some localities for hay, but since it does not produce as much forage as white sweet clover and there is much less demand for the seed, it constitutes only a very small percentage of the total acreage. The much larger root growth of the white species, as illustrated in [figure 5](#), is desirable because of the additional quantity of hummus added to the soil.

The seeds of the yellow species may usually be distinguished from those of other species, as some of them are slightly mottled with purple. The shape of the calyx, which is generally present on unhulled seed, and the venation of the seed pods also distinguish it. (See [fig. 1](#).)

ANNUAL YELLOW SWEET CLOVER.

Annual yellow sweet clover, more commonly known as sour clover or bitter clover, is found chiefly in the South and Southwest. This plant is considered a noxious weed in grain fields throughout the Southwest. It is claimed that the flavor of the seed which is imparted to wheat can not be removed. Bakers decidedly object to this flavor, stating that it injures bread. Sour clover is grown rather extensively as a green-manure crop in

orchards in portions of Arizona and southern California and when properly handled in these regions it has given profitable results. As the seed is obtained from the screenings of wheat, it is offered on the market at a very low price. Occasionally it is sold for the yellow biennial sweet clover. Seed of this plant should not be sown in any part of the United States except the extreme South or Southwest, and then only as a green hay manure crop. Where it is desired to plant sweet clover for pasturage or for the biennial white or biennial yellow species should be used.

OTHER SPECIES OF SWEET CLOVER.

Thirteen species of sweet clover have been tested by the Office of Forage-Crop Investigations to determine their economic value. With the exception of white sweet clover, yellow biennial sweet clover, and yellow annual sweet clover, but four species in the somewhat limited tests have given sufficiently good results to merit special attention, and none have so far proved superior to white sweet clover, which is now extensively grown in many States.

A species of *Trigonella*^[4] is often referred to as blue-flowered melilotus or blue-flowered sweet clover. While this plant is closely related to the plants belonging to the genus *Melilotus*, it does not belong to this genus and therefore should not be called sweet clover. It is an erect, quite leafy, very fragrant annual, which produces a fair growth. It may prove of value as a green-manure crop or as a catch crop under certain conditions, but at the present time it is not to be recommended where sweet clover can be grown successfully. In most tests *Trigonella* has produced less forage than the better species of sweet clover.

^[4] *Trigonella caerulea*.

HISTORY.

Sweet clover has been used as a honey plant and for forage and green-manure for more than 2,000 years in the Mediterranean region, although it has never been considered of much importance.

The first authentic report of sweet clover in the United States was in 1739, when Gronovius stated in his *Flora Virginica* that it was collected by Clayton. Cutler reported its presence in New England as early as 1785, and Pursh in 1814 stated in his *Flora Americae Septentrionalis* that it is found on the gravelly shores of rivers from Pennsylvania to Virginia. Elliott reported the presence of yellow biennial sweet clover in his *Sketch of the Botany of South Carolina and Georgia* in 1824, and Beck found the species *Melilotus leucantha*^[5] in the Northern States in 1833.

[5] Undoubtedly meaning *Melilotus alba*.

In 1856 Prof. Tutwiller, of Green Springs Academy, Ala., received a small quantity of white sweet-clover seed from the secretary to the United States consul in Chile. Part of this seed was planted by a young man named Stendwick on his father's plantation on the prairie limestone belt, where it flourished. This plantation later became the property of J. T. Collins who, realizing the value of this plant, sold seed to persons in many States. Not until recently has sweet clover been grown to any extent as a cultivated crop in this country.

DISTRIBUTION.

While sweet clover is to be found growing in many countries and on all the continents of the world, it is native to temperate Europe and Asia as far east as Tibet. It is grown to a limited extent in England, while in the eastern part of Scotland a small quantity is considered valuable in hay on account of its agreeable odor. The famous Cruyere cheese of Switzerland owes its flavor to yellow sweet clover. In Germany it has given very good results when used as a green-manure, while in parts of Russian Poland and Austria-Hungary it is grown as a green-manure, pasturage, and hay crop on poor soils. This plant is used for forage and as a soil-improving crop in the central provinces of India, while sour clover, commonly referred to as *Melilotus parviflora*, is credited with furnishing 75 per cent of the feed for the cattle of King Island, Tasmania, which produce the best beef and butter sold on the Tasmania market.

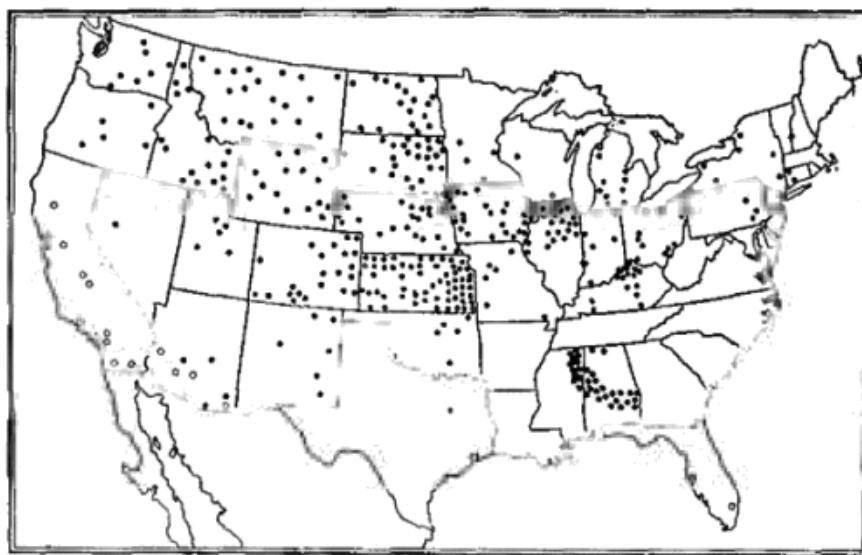


FIG. 6.—Outline map of the United States, showing the localities where sweet clover is grown for forage or for green-manure. Each dot or circle indicates a

county where 50 acres or more is grown under cultivation. The solid dots represent white or yellow sweet clover; the circles represent annual yellow sweet clover.

At the present time sweet clover is grown rather extensively as a field crop in the limestone regions of Alabama, Mississippi, and Kentucky, in northern Illinois, and throughout the western North-Central and Mountain States ([fig. 6](#)): in fact, it is grown as a cultivated crop to some extent in nearly every State in the Union. Comparatively little sweet-clover seed is sown in the Atlantic Coast States, since there the soils are for the most part acid, and heavy applications of lime will be necessary before sweet clover can be grown successfully. It is questionable whether this plant will ever be of much importance in the South Atlantic States, as cowpeas, soy beans, and crimson clover will make a fair growth on those soils in their present condition. The acreage of sweet clover probably will increase in the New England States, where it should prove of value as pasturage and as a soil-improving crop on soils where red clover no longer can be grown. Sweet clover grows abundantly in the limestone regions of northwestern New York.

A much larger acreage of sweet clover is grown in northern Illinois than in any other of the eastern North-Central States. The conditions in the western North-Central States and in the Mountain States appear to be particularly adapted to this crop. It is in that part of the country that the largest acreage is found, and, with the exception of the limestone regions of the South, that the least difficulty is experienced in obtaining a stand.

In those parts of the Mountain and Pacific Coast States, especially Utah, where it has not been tested carefully or where red clover or alfalfa can be grown successfully, sweet clover is looked upon as a weed. It may rightly be considered a weed in the irrigated regions of the West and Northwest, where it grows luxuriantly on ditch Banks. The dissemination of this plant in all parts of the country has been hastened by beekeepers who have seeded it in waste places for the production of honey.

CLIMATIC ADAPTATIONS.

Sweet clover is adapted to a wider range of climatic conditions than any of the true clovers and possibly alfalfa; in fact, it may be grown successfully in any portion of the United States except, perhaps, Florida, and in Florida trials with biennial yellow sweet clover, annual sweet clover, and *Melilotus suaveolens* have been successful. Apparently neither the high temperatures of the South nor the cold winters of the North severely affect the plants, provided there is sufficient moisture in the soil. Comparatively little winterkilling is experienced in Montana, Wyoming, Idaho, and North Dakota when the seed is sown in close drills or broadcasted. Although approximately 50 per cent of the

sweet clover seeded in rows 3 feet apart at Moccasin, Mont., was killed by the unusually severe winter of 1915-16, no winterkilling was noted in plats seeded in close drills.

Sweet clover thrives in the more humid parts of the country, as well as in the semiarid regions where the rainfall is but three-fifths of that required for the normal growth of such crops as red clover and timothy. In the semiarid regions of the West sweet clover has proved to be somewhat more drought resistant than alfalfa.

REQUIREMENTS FOR OBTAINING A STAND.

The requirements for obtaining a stand of sweet clover are somewhat exacting. It is for this reason that so many failures have been experienced. It must not be assumed, because sweet clover is found growing luxuriantly in many waste places and on uncultivated land, that a stand may be obtained by planting it at any time of the year, in any manner, and under all conditions. Throughout the eastern and southern portions of the country, with the exception of a few regions rich in limestone, much care must be used in the preparation of the seed bed, the selection of seed, and the manner of seeding if success is to be expected. For this reason it is necessary to understand fully the requirements for obtaining and maintaining a successful stand.

SOILS SUITABLE FOR SWEET CLOVER.

Sweet clover thrives on the adobe and granitic soils of the Pacific coast; upon the gumbo, hardpan, prairie, and sandy soils of the western North-Central States; and upon the heavy clay, loam, limestone, and sandy soils of the South and East. In fact, it has been grown successfully on all the principal soil types of the United States where the soils were not acid and were well inoculated. It grows luxuriantly on the Selma chalk (rotten-limestone) soils of Alabama and upon soils rich in calcium carbonate in many parts of the country where the lack of nitrogen and humus has caused large numbers of farms to be abandoned. The plants thrive on newly exposed heavy clay soils and upon steep embankments where little else will grow. Sweet clover is more tolerant of poor drainage, overflow, and seepage conditions than alfalfa. In irrigated sections, especially where the reservoir system is in use, large bodies of land are likely to become useless for the growth of alfalfa because of the rising of the water table. On such areas sweet clover will make a vigorous growth. However, maximum growth is to be expected only on well-drained soil.

Sweet clover will do well on many soils which are not fertile enough to grow red clover or alfalfa, and it is on these soils that it will prove most valuable. Like many other plants, it makes its best growth on fertile soils rich in calcium carbonate, although

it will make sufficient growth on poor soils which are not acid to warrant planting it on them. Many hilly pastures may profitably be seeded to sweet clover. It will not only make a valuable addition to the forage of these pastures but will improve the soil so that grasses will grow more abundantly. Some of the best pastures in the Middle West are composed of bluegrass, timothy, and sweet clover.

RESISTANCE TO ALKALI.

Sweet clover grows successfully on soils in the West which apparently are too alkaline for grains or alfalfa. The Wyoming Agricultural Experiment Station reports that it has obtained good yields of sweet clover on seepage land which is so strongly alkaline that no other plants except some of the native grasses will survive, while the California Agricultural Experiment Station found that sweet clover will withstand alkali to a remarkable degree. Prof. F. S. Harris, agronomist of the Utah Agricultural Experiment Station at Logan, claims that it is one of the most alkali-resistant crops grown in Utah, and that in 1913 and 1914 quite an industry developed in some parts of that State in growing sweet clover for hay and seed on land too alkaline for other crops.

In reply to a circular letter on the culture of sweet clover, approximately 100 county agents and extensive growers of this crop located in many parts of the West state that this plant is one of the most alkali-resistant plants grown in their respective districts. In Crook County, Oreg., a good stand was obtained from April seeding in 1915 on a 20-acre demonstration field of sandy loam bottom land so strongly alkaline from black alkali that only salt grass was growing on it before it was planted to sweet clover. This field pastured from 18 to 28 head of calves, cows, and horses from June 1 to October 1 without being irrigated. Sweet clover generally will grow on soils where salt grass^[6] will survive, and it is very much superior to this grass as pasture. After the drainage of water-logged land on which there is a surface accumulation of alkali, it is the common practice in parts of Utah to grow sweet clover for several years before planting alfalfa. It is often stated that alkali land will grow less tolerant crops after sweet clover has been grown on it for a few years. The long roots will open up the subsoil and cause better drainage, thereby affording an excellent means for removing the salts from the soil, as they are readily soluble in water.

^[6] *Distichlis spicata*.

NEED OF LIME ON ACID SOILS.

Sweet clover, like many other legumes, requires a soil containing an abundance of limestone if a maximum growth is to be expected. Throughout the world it makes a luxuriant growth only on calcareous soils. On the black prairie limestone soils of Alabama and Mississippi it grows luxuriantly, although in this region it is very seldom

found on the outcroppings of red clay, which are acid. The distribution corresponds sharply with the line of demarcation between the black prairie soils and other soil types. In some places sweet clover makes a vigorous growth on the Selma chalk (rotten-limestone) soils, while none is to be found on red post-oak clay but a few yards away; yet sweet clover will grow on the red post-oak clay after the soil has received an application of lime. It will thrive on the bald lime-rock spots and rotten-limestone hills of Mississippi, which are so barren that practically no other plants will survive. Thus it appears that lime is essential for the maximum growth of sweet clover in this Region.

The reason for the exceptional growth of sweet clover in north-central Kentucky is undoubtedly the fact that these soils contain an abundant supply of limestone. The Kentucky Agricultural Experiment Station states that this area is the only portion of the State where sweet clover is being grown with general success without applying lime.

Soils on which sweet clover is aggressive are almost invariably alkaline or but slightly acid. This plant is often found in valleys of streams in localities where the soils are supposedly acid, but such streams generally have their origin in limestone areas or flow through limestone regions, and calcium carbonate is thus deposited in these valleys during flood periods with the sedimentary deposits from flood waters. Sweet clover often appears in deep cuts along highways or railroads in localities where the soil is known to be acid and where sweet clover has not previously grown. In many of these cuts the acid soil has been removed and neutral or alkaline subsoil exposed, or limestone has been used in ballasting or road making and the dust has blown on the exposed soil. It is a very common occurrence to find sweet clover making an abundant growth along macadamized roads from which the wind has scattered the finely pulverized limestone.

An application of burnt lime or finely ground limestone has made the difference between success and failure in most experiments which have thus far been conducted on decidedly acid soils. ([Fig. 7.](#))

A number of sweet-clover experiments were performed on acid soils and on adjacent plats or fields of the same type of soil that had received applications of limestone varying from 1 to 4 tons to the acre. There was a marked difference in the stands obtained and in the growth of the plants on the limed and unlimed areas. In some cases the difference in growth was so marked that the last round of the lime spreader could be distinguished at some distance from the plats. The stands were much heavier on the limed areas and the plants made from two to three times more growth than those on the unlimed plats. Yields of hay were doubled on soils that received only sufficient limestone to neutralize the acids in the surface soil, although the yields were further increased when more limestone was added.

Mr. W. E. Watkins, county agent of Allen County, Kans., made counts of the number of plants which winterkilled during the winter of 1914-15 on given areas of

limed and unlimed soil. It was found that from 15 to 35 per cent more plants winterkilled on the unlimed soil than on the limed areas. That portion of the unlimed field on which the fewest plants winterkilled was found to have the lowest lime requirement. On the unlimed areas with a low lime requirement 15 per cent more plants winterkilled than on the limed areas; on those with a high lime requirement the increase in winterkilling was 33 per cent. In the fall of 1914 the hay cut from the limed areas exceeded that from the areas with a low lime requirement by 600 pounds per acre and exceeded that from the areas of high lime requirement by 4,000 pounds per acre. In July, 1915, the increase in hay yield on the limed areas over that from the areas with a low and with a high lime requirement was 2,300 and 9,400 pounds per acre, respectively. The area of high lime requirement returned a small yield in 1914 and no hay in 1915.

In spite of the fact that sweet clover is as sensitive to soil acidity as red clover or alfalfa, a large percentage of the acreage thus far seeded in the eastern half of the United States has been composed of acid soils, and this soil acidity undoubtedly is responsible for a very large percentage of the failures with sweet clover in this section. Where sweet clover is to be sown on acid soils a sufficient quantity of lime should first be applied to at least neutralize the fields in the soil to a depth of 6 inches. An application of 1 ton of burnt lime or 2 tons of finely ground limestone will usually be sufficient for this purpose.

Fields have been noted where sweet clover was making a fair growth on apparently acid soils. Such fields usually are rich in humus or phosphorus and are exceptional cases rather than the rule.

Soil types which have slightly acid surface soils and alkaline subsoils will grow sweet clover successfully, provided the acid soil is not more than 6 to 12 inches in depth.

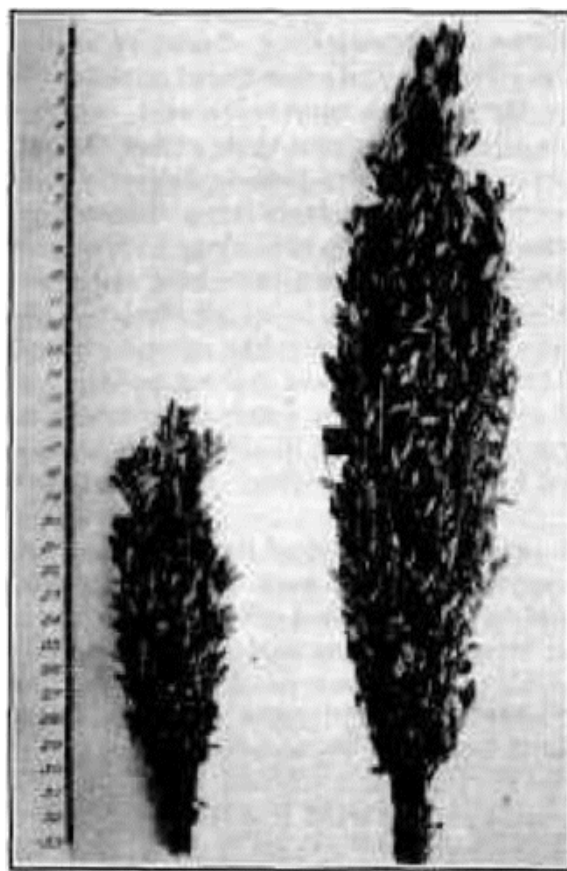


FIG. 7.—Sweet-clover plants, showing the effect of lime upon their growth. The plants at the left represent the average growth on the unlimed portion of a field; the plants at the right show the average growth on the limed part of the same field.

FERTILIZERS.

Owing to the fact that sweet clover thrives on the barren Selma chalk (rotten-limestone) hills of Alabama and Mississippi and grows abundantly on worn-out, abandoned land in north-central Kentucky, it is often assumed that it will grow on soils too depleted in plant food to produce other crops. These regions represent soils which have become exhausted primarily in nitrogen and humus as the result of continuous cropping with nonleguminous plants. Some of these soils contain sufficient phosphorus and potassium for fair crop production, although this supply may be in such a condition that it will not become available fast enough to supply the needs of most crops. Sweet clover, like all legumes, has the power to extract nitrogen from the atmosphere, and on account of its extensive root system it is able to obtain phosphorus and potassium from a larger area than most plants. The large roots not only add a quantity of humus and nitrogen to the soil but they also open it up to a considerable depth, thus providing better aeration and improving its physical condition. Improved physical condition causes the bacterial flora to increase and thereby indirectly causes a larger quantity of unavailable phosphorus and potassium to be made available for plant use.

On soils which are known to be low in phosphorus or potassium an application of fertilizer containing the necessary element should be made when sweet clover is sown without a nurse crop. However, when it is sown with a nurse crop or in the late summer or early fall on grain stubble, the residues left in the soil from fertilizers applied to the nurse crop will, under ordinary conditions, be sufficient for the plants. That sweet clover will respond readily to applications of phosphorus on soils low in this element has been well demonstrated by the farmers of Livingston County, Ill. In this county finely ground rock phosphate was applied to a portion of a number of fields at the rate of 1,500 to 2,000 pounds per acre. The phosphate was thoroughly incorporated with the soil just before seeding oats and sweet clover. In the growth of sweet clover there was a marked difference the following year between the treated and untreated portions of the fields. Those portions of the fields which received an application of phosphate not only contained many more plants on a given area, but the vigor and growth of the plants were most marked. On June 1 the plants on the treated areas were 12 to 15 inches taller than those on the untreated parts of the fields. This difference in the thickness of stand and the height of plants was so striking that the last round of the phosphate spreader was plainly distinguishable.

Yields of sweet-clover hay have been increased as much as 2 tons per acre from applications of barnyard manure. Such an increased yield would be equal approximately to 8 tons of green-manure. Some people may consider it poor farm practice to apply manure to such crops as sweet clover, but it is very probable that the cumulative effect of the increased yields of the following crops, especially on soils low in organic matter, will be greater than if the manure is applied to other crops. Heavy applications of manure to the preceding crop should also greatly benefit sweet clover.

USE OF A NURSE CROP.

If sweet clover is to become an important crop throughout the North-Central States it must necessarily be seeded with grain. Good success has been obtained by seeding sweet clover in the spring on winter grain or with spring grain on soil that was inoculated and not acid. Seed may be broadcasted in the early spring on winter grain when the ground is in a honeycombed condition, or it may be sown later when the ground may be cultivated. A large acreage of sweet clover is sown in the western North-Central States and in Illinois in the spring with oats, barley, or wheat as a nurse crop. Early varieties of oats and spring wheat have given somewhat better results in portions of the Northwest than barley. In Illinois oats are used almost entirely. Only a few fields were noted where flax had been used as a nurse crop, but in these fields it was successful. In wet seasons the sweet clover may make a growth sufficiently large to interfere seriously with harvesting the flax. On this account this combination should be tested thoroughly in an experimental way before being recommended for general field practice.

In those sections of the country where the moisture supply is limited, sweet clover should be sown without a nurse crop. Failure to obtain a stand is more likely to occur when the seed is sown with grain than when it is sown alone, because during dry weather, which is likely to occur when the grain is maturing, the supply of moisture in the soil is apt to be insufficient for both crops. When this condition prevails the clover will suffer badly and in some cases be killed. When sweet clover is sown with a nurse crop it is strongly recommended that the grain be seeded at not more than two-thirds the usual rate. This will give the sweet clover a much better chance than when a full seeding is made. When severe droughts occur it may be necessary to cut the grain for hay if the stand is to be saved.

CHOICE OF SEED.

On account of the low germination of much of the sweet-clover seed offered for sale it is very important that seed be tested for germination before planting. Low germination usually is due to the fact that many of the seeds remain hard after they have been in the germinator or soil for a month or more. The seed coats of hard sweet-clover seeds become permeable to water very slowly, if at all, in storage. The germination of such seeds is greatly increased, however, when they are subjected for a time to alternating temperatures, such as freezing and thawing. It is on this account that unhulled seed, which germinates poorly in the laboratory, often will produce good stands when sown during the winter. When sweet clover is to be sown in the spring it is very important that only hulled seed which germinates 75 per cent or more be sown. As explained later under the heading "Seeding," unhulled seed which has a low germination should be used for seeding only during the winter months, so that there will be sufficient time for the alternating temperatures of winter and early spring to cause it to germinate during favorable weather.

Hulled seed usually germinates much better than unhulled seed, as is shown in Table I.

TABLE I.—*Germination and hard seed content of samples of sweet-clover seed, hulled, and unhulled, from different sources.*

Description.	Number of samples.	Average percentage of —	
		Germination.	Hard Seed.
Kind of seed:			
Hulled	237	53.25	18.7
Unhulled	45	11.8	70.9
Source of seed:			
Southern	22	14	60
Northern	22	37	43
Imported	28	56	12

Table I shows that northern-grown seed germinates better than northern-grown seed and imported seed better than either. The low germination of the southern-grown seed is probably due to the fact that a very large percentage of it is flailed out and sown in the hull. Northern-grown seed generally is thrashed with either a grain separator or a clover huller. Imported seed always is hulled. In hulling seed the rasps or concaves of the machines scratch the seed coats sufficiently to permit water to penetrate them, so that the germination is greatly increased. Apparently there is no reason why southern-grown seed when it is properly hulled should not germinate as well as northern-grown seed.

Since the Ames scarifying machine ([fig. 8](#))^[7] has been placed on the market, it is possible to buy scarified seed. This machine is so constructed that the seed is forced through a conveyor, part of which is covered with sandpaper. When the seed comes in contact with the sandpaper it is scratched, so that water will penetrate the seed coats. When this machine is run properly the germination of seed is greatly increased, but when carelessly operated germination may be lessened, as many of the seeds may be broken.

^[7] This machine was invented by Prof. H. D. Hughes, of the Iowa State College at Ames. A United States patent covering this device has been issued and dedicated to the free use of the public.

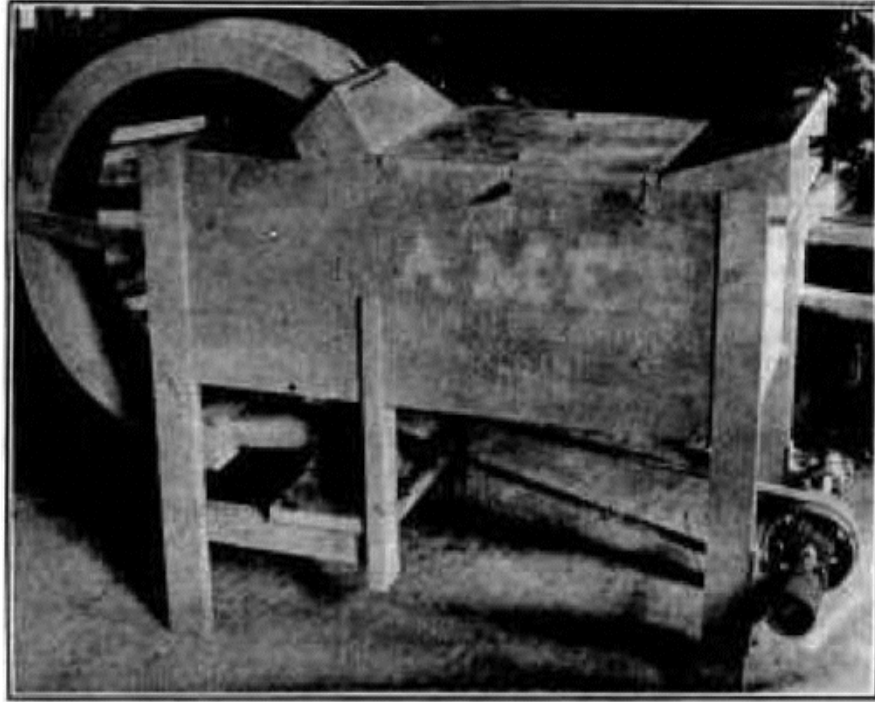


FIG. 8.—Ames hulling and scarifying machine.

The retarded germination of sweet-clover seed may be overcome by soaking it in commercial concentrated sulphuric acid for 20 minutes. It should then be washed quickly, using running water if possible, as sulphuric acid becomes very hot when mixed with small proportions of water. A great deal of water therefore is necessary in order to lessen the danger of burning. The seed should be dried quickly by spreading it out on a floor or canvas, and it should be stirred at intervals. Unhulled seed should never be treated with sulphuric acid. When only a small quantity of sulphuric acid comes in contact with the hulls a very high temperature will result and the seed will be killed. The treatment of seed with sulphuric acid for seeding on a field scale is not to be recommended, in view of the fact that as good or better results may be obtained by using scarified seed.

It is very important that seed of the desired species be obtained. Many lots of sweet-clover seed offered for sale on the market consist of mixtures of the yellow and white species, and many samples also are adulterated with alfalfa. Seed which is simply labeled sweet clover should never be purchased, as seed so labeled may be any one of the several varieties offered for sale. It is always best to state the specific kind of seed ordered and then submit a sample to either your State Agricultural experiment Station or one of the seed laboratories of the United States Department of Agriculture^[8] for identification before purchasing.

^[8] Samples of seed may be submitted for analysis or identification to the Seed Laboratory of the United States Department of Agriculture at Washington, D. C., or to any of the following laboratories maintained through the cooperation of the Department: Branch Seed-Testing Laboratory, Agricultural Experiment Station, Columbia, Mo.; Branch Seed-

Testing Laboratory, Agricultural Experiment Station, Baton Rouge, La.; Branch Seed-Testing Laboratory, Oregon Agricultural College, Corvallis, Oreg.; Branch Seed-Testing Laboratory, Purdue University, La Fayette, Ind.; Branch Seed-Testing Laboratory, California Agricultural Station, Berkeley, Cal.

The Seed Laboratory of the United States Department of Agriculture during the winter of 1915-16 obtained 172 trade samples of sweet-clover seed and, as may be seen from Table II, many of the samples were not true to name.

TABLE II.—*Trade samples received in response to requests for white sweet clover seed.*

Seed labeled—	Number of samples.	Seed when tested found to be—						
		White sweet clover.	Biennial yellow sweet clover.	White and biennial yellow sweet clover.	Annual yellow sweet clover.	Alfalfa and—		
						White sweet clover.	Biennial yellow sweet clover.	White and biennial yellow sweet clover.
White flowering sweet clover, white sweet clover, or Holhara clover, or <i>Melilotus alba</i> .	147	91	10	28	—	13	1	4
Sweet clover	22	6	4	2	5	5	—	—
Not labeled	3	1	—	—	2	—	—	—
Total	172	98	14	30	7	18	1	4

PREPARATION OF THE SEED BED.

Sweet clover requires a well-settled and firm seed bed, with just sufficient loose soil on the surface to permit the seed to be well covered. When the seed is sown in the spring on winter grain the seed bed usually is in good condition. At this season of the year the seed may be sown, so that it will be covered by freezing and thawing weather. It may be sown also when the ground is in condition to cultivate and then may be harrowed or drilled in. When the seed is sown with spring grain the seed bed is not as firm as it should be for the prompt germination and establishment of the young clover plants. If sown in this manner the soil should be worked into a fine condition and firmed as much as possible. It is good practice to roll the ground with a corrugated roller after seeding. Better stands are usually obtained by seeding on fields that have been disked and harrowed than on those that have been plowed.

When sweet clover is seeded without a nurse crop it should not be sown on freshly plowed land which has had no opportunity to settle. The land preferably should be plowed several months before the seed is to be sown, and then worked at intervals with

soil packers or harrows. Double disking and harrowing just previous to seeding are to be strongly recommended in preference to plowing at this time.

When sweet clover is to be seeded in the fall on grain stubble, the ground should be disked and worked into good condition as soon as the grain can be removed. If the seed is sown immediately the field should be rolled after seeding.

Fall-plowed ground ordinarily makes an ideal seed bed for spring seeding. Soil which has been previously planted to a cultivated crop, such as corn, is usually put in sufficiently good condition for sweet clover by disking. Good success has been attained by merely broadcasting the seed on sandy soil and scratching it in with a harrow. Such a seed bed appears to be ideal when the seed can be covered sufficiently to insure plenty of moisture. It must be remembered that young sweet-clover plants are not drought resistant and that every precaution should be taken in seasons of drought or on land which drought affects badly to so prepare the seed bed that the largest quantity of moisture will be conserved.

Excellent stands have been obtained at times by double-disking native prairie sod and either covering the seed with a harrow or sowing it with a drill.

SEEDING.

The proper time to seed sweet clover should be determined by the germination of the seed, the climatic conditions of the region, and the condition of the seed bed at the time of sowing. When growing under natural conditions, seed which has lain in the ground over winter germinates in sufficient quantity during the following spring to produce a stand. It is therefore assumed that since this seed has passed the winter on or in the ground and has produced a good stand the following spring, sweet clover may be sown at any time of the year and a satisfactory stand obtained. Little is thought of the enormous number of seeds which shatter from a single plant and fall on an area not exceeding 5 or 6 feet in diameter. Single plants have produced as many as 350,000 seeds (the approximate number in 1½ pounds), or about 10,000 seeds for each square foot of ground covered. It matters little how many of these seeds germinate in the fall they mature or during the following winter, when the seedlings will be killed by freezing, for there will be enough viable seeds left in the ground to germinate when conditions are favorable in the spring. Conditions are very different when sweet clover is sown on cultivated soil at the rate of 5 to 20 pounds of seed to the acre—25 to 100 seeds to the square foot. When this quantity is sown, it is necessary that it be planted at such a time that the greatest number of seeds will germinate and produce plants.

HULLED SWEET-CLOVER SEED.

Hulled seed makes up a large percentage of the sweet-clover seed sown. The germination of hulled seed varies considerably, although ordinarily it is higher than that of unhulled seed. Seeding experiments conducted at Arlington, Va., with seed which germinated 80 per cent show clearly that seed which germinates well should not be sown during the winter months in those sections of the country where midwinter thaws are likely to occur, and especially in sections south of the latitude of southern Ohio. In these experiments seed was sown during each month of the winter. Good stands were obtained only on those plats which were sown in the latter part of February and during March and April. At least 75 per cent of the seed sown during November, December, and January germinated on warm days during winter thaws and was killed by later cold weather. Notwithstanding the fact that sweet-clover seedlings will endure fairly low temperatures, seed germinating more than 50 per cent should not be sown during the winter months, and preferably not more than a week previous to the average date for the last severe freeze.

No data have been secured on winter seeding in those portions of the United States where open winters do not occur. It is probable that in those sections the winters are sufficiently cold to prevent germination before spring. Good results may be obtained by winter seeding, but as usually no trouble is experienced in those sections in obtaining a stand by seeding as soon as the soil can be worked in the spring, it is strongly recommended that seeding be done with hulled seed, which germinates well at this time of the year.

Many excellent stands have been obtained by seeding late in the spring, but in most sections seeding at this time is not as certain to produce a good stand as earlier seeding. Late spring seeding may be preferable when the ground is weedy and the clover is to be seeded without nurse crop. Under these circumstances a crop of weeds may be destroyed before seeding.

Very good success has been obtained in the Southern and Central States, and in some of the Northern States, by seeding sweet clover in the late summer or early autumn. When there is sufficient moisture in the soil for germination and when good seed is used, better stands have been obtained by seeding about eight weeks before severe frosts are to be expected than from spring sowing. This is particularly true in regions where late spring droughts or severe summer droughts are likely to occur. Seeding at this time may be done after an early crop has been harvested and when weeds are not likely to be troublesome. Plants from fall seeding mature from 10 days to two weeks later the following season than plants from spring sowing of the same year. The later time of maturing is an advantage, in that the plants will be ready to cut during better haying weather. The root growth is not as large from fall seeding as from spring seeding, and therefore not quite as much humus is added to the soil. Late fall seedings are very likely to be injured from heaving on wet clay soils.

UNHULLED SWEET-CLOVER SEED.

Unhulled sweet-clover seed is sown principally in Kentucky, Alabama, and Mississippi. On the limestone soils of regions, which appear to be naturally adapted to sweet clover, very good results are obtained by using unhulled seed. It is not because southern-grown unhulled seed germinates better than northern-grown unhulled seed that better stands are obtained in the South from it, but it is mainly because southern farmers better understand the somewhat exacting conditions necessary for obtaining a stand with this kind of seed. Unhulled sweet clover contains a large percentage of hard seeds which will not germinate until they have been in the soil for some time and have been subjected to varying temperatures.

Seeding experiments have been conducted at Arlington, Va., where unhulled seed which contained 90 per cent of hard seed was sown during each month of the winter. Good stands were obtained on those plats seeded at the rate of 24 pounds (3 pecks) of seed to the acre during December and January, and fair stands on the plats seeded at this rate in February. Later seedings failed to produce a stand.

A large percentage of the unhulled seed sown in the South is seeded during January and the first part of February. Good stands are seldom obtained from unhulled seed south of the latitude of Washington, D. C., when the seed is sown later than the middle of February.

The use of unhulled seed has usually been attended with failure in the northern portion of the United States, although occasionally good stands have been obtained the following spring from late fall seeding. This failure is in part due to the fact that the seed has been sown in the spring and at a time when only seed germinating well should be used. When unhulled seed is to be sown north of the latitude of Washington, D. C., it should be sown not later than February 15, and preferably earlier. Observations show that fairly good stands may be obtained by seeding during the winter, but care should be taken not to sow seed earlier than necessary on land which is subject to washing. Farmers should have no trouble in purchasing hulled seed, and therefore it is recommended that only hulled seed which germinates well be sown.

RATE OF SEEDING.

The rate at which sweet clover should be seeded varies with the germination of the seed, the condition of the seed bed, the climatic conditions of the region, and the method of seeding. Throughout the humid sections of the eastern United States sweet clover ordinarily is seeded at the rate of 15 to 20 pounds of hulled seed to the acre. From 12 to 15 pounds should be ample where the seed bed is in good condition and the seed germinates 75 per cent or more. In Illinois, the western North-Central States, the Mountain States, and the Pacific Coast States good stands are generally obtained by sowing 10 to 12 pounds of hulled seed to the acre. In eastern Washington it is claimed that from 5 to 8 pounds to the acre are sufficient for good stands.

When sweet clover is grown under irrigation, 8 to 10 pounds of hulled seed usually are sufficient, and from 2 to 4 pounds per acre are enough when seeded in rows from 2 to 4 feet apart. Of unhulled seed 3 to 6 pecks (24 to 48 pounds) or 20 pounds of hulled seed are usually sown in the South for pasturage or hay. In any region at least 10 pounds more of the unhulled than of hulled seed should be sown to an acre. Unless annual yellow sweet-clover seed is thoroughly cleaned it should be sown at the rate of 25 to 30 pounds to the acre.

METHODS OF SEEDING.

The methods used for seeding red clover or alfalfa in any particular region will be suitable for seeding sweet clover. Good results have been obtained by broadcasting the seed on winter grain in the spring when the ground is in a honeycombed condition. Perhaps a better method is to wait until the ground can be worked and then to broadcast and cover the seed with a harrow or to sow it with a drill. Unhulled seed is usually broadcasted, since it is necessary to sow it before the ground is in condition to be worked. Unless the hulls have been rubbed smooth, some difficulty may be experienced in seeding it evenly with a drill.

When sweet clover is to be sown with spring-seeded grain or when it is to lie sown without a nurse crop it may be drilled in or sown broadcast and covered with a harrow. Better stands are generally obtained with a smaller quantity of seed when it is sown with the drill than when it is broadcasted on honeycombed ground. When the seed is sown at the time the grain is planted, the grass-seeder attachment of the drill commonly is used. In some sections the end-gate seeder is used almost entirely. When the seed is sown by either of these methods it may be seeded alone or mixed with the grain. When only the clover seed is sown with a drill, the alfalfa and clover seed drills are to be preferred.

Sweet-clover seed may be mixed with some inert substance of approximately the same size and weight and sown with an ordinary grain drill. Finely cracked corn, cracked wheat, or coarse bran often are used for this purpose. When one portion of sweet clover is mixed with two portions of a filler find the drill is set to sow one-half bushel of wheat, it will usually sow from 15 to 20 pounds of sweet clover to the acre. As this quantity will vary with the different types of drills, it is necessary to test each drill, so that the seed may be mixed with the filler in such proportions that the desired quantity will be sown. The drill may be tested by blocking it up, so that the geared wheel is off the ground, and this wheel may be turned a sufficient number of times to establish a definite portion of an acre. The seed that runs through can then be weighed and the rate per acre determined. The rate may be determined more accurately by plugging up the grain tubes or by tying a small sack on each tube and pulling the drill for a specific distance over the field to be sown. The jar of the drill will cause it to drop more seed than when it is blocked up and run by hand.

It is often desired to seed sweet clover on land which can not be cultivated. When sown on such land it is recommended that unhulled seed or seed that contains a large percentage of hard seed be used and that it be broadcasted during the winter. The subsequent freezing and thawing will cover many of the seeds and cause them to germinate. It is a good plan to scatter in deep gulleys mature plants that have not shattered all their seed. The branches of these plants will help to hold the seed in place until it germinates and the young plants become established.

Seed may be scattered on native prairie ground in the late winter, but unless it is trampled into the ground by live stock disappointing results are likely to be obtained at first. Fair results have been secured by planting seed with disk drills on prairie sod after it had been double-disked in the early spring. This method should be used in preference to broadcasting the seed and depending on cattle to trample it in. Mr. George Hummer, of Prairie Point, Miss., reports good success in his locality by simply broadcasting 1 peck of unhulled seed on Bermuda-grass sod not later than January 1.

INOCULATION.

Excepting soil acidity, lack of inoculation probably is responsible for more failures with sweet clover than any other one cause. When sweet-clover plants are not inoculated they must depend upon the available nitrogen in the soil for their supply, and as the crop is grown for the most part on soils low in nitrogen the plants can not be expected to make more than a small growth. ([Fig. 9.](#))

Army and Thatcher, at the Minnesota Agricultural Experiment Station, obtained 10 times as much dry matter in the tops and seven times as much in the roots of sweet-clover plants which had been grown on thoroughly inoculated soil as from plants which had been grown on soil not inoculated. Moreover, the plants grown on the inoculated soil contained 117 pounds more nitrogen to the acre than those grown on the uninoculated soil.

Experiments in many other sections of the country, and especially in the northeastern quarter of the United States, where but little sweet clover or alfalfa has thus far been grown, show that inoculation is very essential to success. Ordinarily it is not necessary to inoculate sweet clover when it is to be planted on land where alfalfa, bur-clover, or black medic thrives, because the same strain of inoculating germs inoculates all of these plants. However, when this clover is to be planted on land where none of the plants inoculated by this strain of the organism have been grown, inoculation should be provided. In localities where sweet clover or other plants inoculated by the same strain of bacteria thrive, the early growth has at times been made much more vigorous by inoculating the soil or seed thoroughly.

It is not safe to assume that a certain piece of soil is inoculated because any one of the plants inoculated by the same strain of the organism is growing or has been grown on other fields in the same vicinity. Many fields have come under observation where sweet clover was a failure because the plants were not inoculated, when plats or fields of alfalfa growing near by were abundantly inoculated.

There are several methods of inoculating sweet clover, any one of which when properly applied should give good results.

THE SOIL-TRANSFER METHOD.

The soil-transfer method of inoculation consists in scattering over the field to be seeded 200 to 400 pounds to the acre of soil collected from sweet-clover, alfalfa, or bur-clover fields where the healthy plants show an abundance of tubercles on the roots. It is strongly recommended that this soil be scattered on a cloudy day or in the early morning or in the evening and immediately harrowed or disked into the ground, as the sun's rays are very injurious to the inoculating germs. It is a good plan for the person scattering the soil to walk directly in front of the harrow. When this practice is followed little harm can be done by the light. To facilitate even scattering, the soil may be mixed thoroughly with two or three times its weight of other soil, preferably from the field where the sweet clover is to be sown. Soil used for inoculating sweet clover does not necessarily have to be scattered on the land just previous to sowing the seed. It may be scattered a few months or a year in advance of the time the sweet clover is to be seeded and be just as effective as if it were scattered at a later time. In general, where sweet clover is to be seeded in the spring on winter grain, the inoculation should be applied before the grain is sown.

Good success has been obtained by drying in a dark place soil containing the inoculating germs, sifting it, and running it through the fertilizer compartment of a

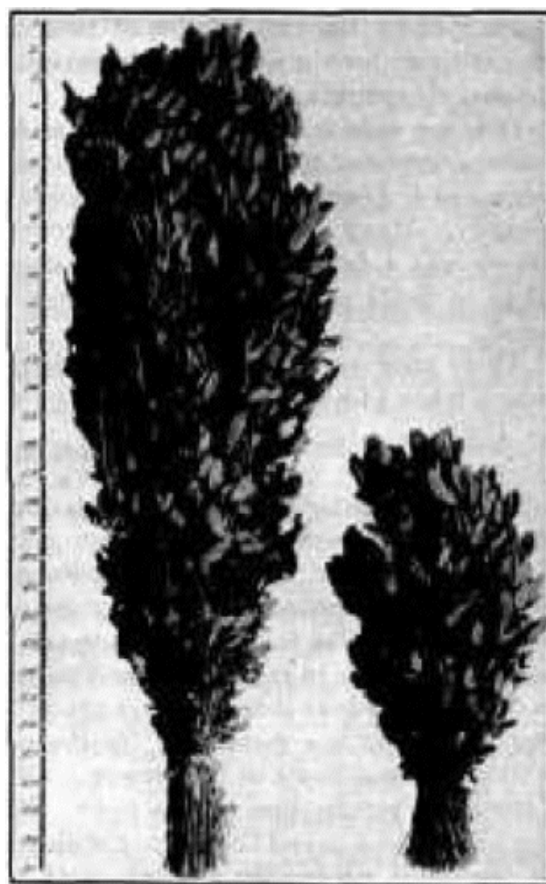


FIG. 9.—White sweet clover at Arlington, Va., showing the effect of inoculation upon their growth. The plants at the left represent the average growth on the inoculated plats; those at the right the average growth on the plats not inoculated. The plats had been previously limed and were seeded on the same date.

grain drill. When this method is employed it is not necessary to use as much soil as when it is scattered broadcast.

A comparatively new method which has given successful results calls for dampening each bushel of seed and spreading it on a cloth, paper, or cement floor, where half a gallon of thoroughly inoculated soil from sweet-clover or alfalfa plants may be sifted over it. Some people prefer to add a trace of glue or sugar to water, so that more of the soil will adhere to the seed, although some soil will remain on the seed if the glue or sugar is not used. When only this quantity of soil is used it should be collected from around the roots of sweet-clover plants which are abundantly inoculated. Such soil may be collected in the fall and kept until spring in a cool, dry, dark place with no injury to the inoculating organisms. Seed treated in this manner should be kept in the dark and should be sown as soon as possible after treating.

THE PURE-CULTURE METHOD.

The pure-culture method has the advantage of greater ease of transportation and freedom from danger of introducing harmful pests upon the farm. Inoculation by pure cultures may be carried out in either of two ways:

(1) A bottle of pure culture of the proper kind of bacteria is opened and the culture mixed with a convenient quantity of water; this diluted culture now is mixed thoroughly with a considerable quantity of soil, preferably from the field where the legume is to be sown; the treated soil is then distributed in the same manner as when inoculation is made by the soil-transfer method.

(2) A pure culture of the proper kind of bacteria should be applied to the seeds in such a way that they will all be moistened. The seed should then be permitted to dry in a shady or dark place and should be planted as soon as possible after it is dry. Drying may be facilitated by adding dry, sifted soil, preferably from the field where the seed is to be sown. Inoculating organisms very often die within a week after the seed is inoculated. It is highly desirable, therefore that the inoculation be made the day the seed is sown. Inoculated seed never should be dried in the sun.

The question is often asked whether it is advisable to inoculate seed with pure-culture method and sow it on honeycombed ground in the spring. No experiments have thus far been completed to determine the advisability of this procedure. Some inoculation would probably result from this practice, because the bacteria on that portion of the seed next to the ground would be protected from the sunlight and would in a short time under ordinary conditions be covered by the freezing and thawing of the soil. While it is hardly possible to obtain as complete inoculation by this practice as by other methods, it is to be preferred to no inoculation.

TREATMENT OF THE STAND.

The manner in which a stand of sweet clover is handled should depend somewhat upon the method and date of seeding and the purpose for which it is sown. Climatic conditions should also be taken into consideration and the handling of the crop governed accordingly.

TREATMENT THE FIRST SEASON.

The most serious objection to seeding sweet clover in the spring without a nurse crop is weeds. In many sections of the country seeds will take as much water from the soil and make as much or more shade than a crop of grain. In spite of the fact that sweet clover will withstand more adverse conditions than red clover or alfalfa, a heavy growth of weeds will greatly retard the growth of the plants and in some cases kill most of them. ([Fig. 10.](#)) On plats sown in April without a nurse crop at Arlington, Va., it was necessary to mow weeds five times during the summer of 1915 in order to keep them partly checked. Where it is necessary to mow a field so many times the plants are not only checked or killed, but as much time is required for this work as would be necessary to harvest a crop of grain. This trouble may be overcome in part by pasturing the sweet clover the first season, but even then during wet weather it may be necessary to cut the weeds at least once before the plants become well enough established to turn live stock on the field. The plants should at no time be clipped closer than 5 inches from the ground.

After a field of sweet clover has become well established, it may be pastured throughout the summer and fall. Close grazing should be avoided during the summer, or the plants may be killed, but they may be pastured fairly close to the ground in the autumn, as it does not appear as necessary to provide a winter covering as is the case with red clover. Close pasturing or clipping late in the fall has had a marked effect on the growth of the plants the following spring on some fields and no apparent effect on the stand and growth of the plants on other fields. A portion of a field in Livingston County, Ill., was clipped close to the ground in the late fall of 1915. On June 1, 1916, the stand was somewhat heavier on the unclipped part of the field. More noticeable than the thickness of the stand was the fact that the plants on the unclipped portion were 8 to 10 inches higher than those on the clipped area. It is reasonable to believe that plants going into the winter with no protection are more likely to be injured than those having some protection.

On the other hand, many fields in different parts of the country have been closely clipped or pastured in the late fall with no noticeable injury. Because of the value of the hay or pasturage in the late summer and autumn of the year of seeding, it is strongly recommended that the first year's growth be utilized. If the field be cut for hay it is well to leave a 4-inch or 5-inch stubble, as this will serve to catch drifting snow during the

winter, thereby adding to the protection against winterkilling. If the field is not pastured the first season and weeds are not troublesome, a cutting of hay may be made when growth ceases in the fall.



FIG. 10.—White sweet-clover plant (at the left), showing the effect of a heavy growth of weeds. Had the weeds not been present the plant at the left should have been larger than the one at the right, as the seed was sown two weeks earlier and the other conditions for growth were ideal. Four-fifths of the plants on the plat which had a heavy growth of weeds were entirely killed.

When sweet clover is seeded with grain, moisture conditions should serve to determine whether the grain should be permitted to ripen or be cut for hay. When untimely droughts appear the plants may be killed if the grain is not cut as early as possible.

In the South and in some sections of the Eastern and North-Central States where the soil contains an abundance of limestone and is well inoculated, a cutting of hay may usually be obtained after a grain crop has been harvested. In other sections of the North

in only exceptionally favorable weather will more than pasturage be obtained after the grain is cut.

TREATMENT THE SECOND SEASON.

One of the special advantages of sweet clover is that it produces good pasturage somewhat earlier in the spring than most forage crops. In the North, with the exception of the extreme northern portion of the United States, it will furnish a cutting of hay in June or excellent pasturage until that time and a crop of hay or seed in late summer. In the South two cuttings of hay and a seed crop may be harvested. After maturing seed the plants die. It is a common practice in many sections to pasture the crop until about June 10, when the stock is removed and the plants are permitted to mature seed. If the plants have not been grazed closely they should be clipped at this time, so that the seed crop will ripen more evenly. Sweet clover may be pastured during the entire second season's growth, provided sufficient stock is kept on the field to prevent the growth from becoming woody. If the plants become coarse the pasture may be clipped, leaving an 8-inch stubble, so as to induce a new growth which will be more palatable. If it is desired to have the pasture reseed itself stock should be removed at least eight weeks before heavy frosts are expected, or only sufficient stock should be permitted to remain on the pasture to keep some of the plants in check.

SWEET CLOVER IN MIXTURES.

Very little sweet clover thus far has been grown in mixtures with other crops. A few farmers have sown red clover and sweet clover together, but such a mixture has no advantage over sweet clover seeded alone for hay, as sweet clover should be cut at least two Weeks before the red clover is ready to harvest. Sweet clover is being seeded to some extent on native prairie sod in the Northwest, where it is claimed it adds greatly to the value of the native grasses for pasturage. A thin seeding of sweet clover is often desired in bluegrass pastures on this account. One of the best pastures in eastern Iowa consists of a mixture of bluegrass, timothy, and sweet clover. The Mississippi Agricultural Experiment Station recommends a mixture of Johnson grass and sweet clover. In this mixture the first cutting will consist of almost pure sweet clover, while the second and third cutting's will be a mixture of those plants. A number of southern farmer have had good success in seeding sweet clover on Bermuda-grass sod.

The New Jersey Agricultural Experiment Station has obtained excellent results from a mixture of Dwarf Essex rape and sweet clover, and also by the addition of soy beans to this mixture. It was found that by seeding 6 pounds of rape and 10 pounds of sweet clover per acre an abundance of nutritious pasturage was produced and that pigs

preferred this mixture to alfalfa. When soy beans were added it was seeded at the rate of 1 bushel of soy beans, 6 pounds of Dwarf Essex rape, and 18 pounds of sweet clover. The soy beans were drilled by themselves, and the rape and sweet clover were mixed and seeded with a press drill. Brood sows made a gain of from three-fourths to 1 pound a day during July on this mixture without additional feed and gave unusual evidence of thrift and vigor.



FIG. 11.—A cornfield, showing the effect of fall and spring plowing in killing sweet clover that had made but one year's growth. The portion of the field at the left was plowed in the autumn, while that at the right was plowed the following spring, after the plants had started growth. The corn is 4 inches high.

ERADICATION OF SWEET CLOVER.

Some farmers hesitate to plant sweet clover on their farms for fear they will have difficulty in eradicating it when the fields are planted to other crops. The results obtained annually by hundreds of farmers are sufficient proof that there is no foundation for such fear; in fact, farmers are experiencing much difficulty in cutting the first crop the second season so high that the plants will not be killed. The new crop of sweet clover, unlike that of red clover and alfalfa, must come from the buds left on the stubble, so when the plants are cut below these buds they will be killed. As sweet clover is a biennial, the plants die as soon as the seed crop is produced.

When the first year's growth of sweet clover is to be turned under for green-manure it is recommended that the field be plowed after the plants have made some growth the following spring rather than in the fall of the year of seeding. When the first year's growth is plowed under the same fall many of the plants will not be entirely covered,

and these will made a vigorous growth the following spring. When the plowing is delayed until the plants have made some growth the following spring no trouble will be experienced in eradicating them. ([Fig. 11.](#))

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